

PVsyst - Simulation report

Standalone system

Project: EME building Corridor Illumination

Variant: New simulation variant

Standalone system with batteries

System power: 4320 Wp

EME building, CUET, Chattogram - Bangladesh



Project: EME building Corridor Illumination

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
02/02/26 19:54
with V7.4.7

Project summary

Geographical Site

EME building, CUET, Chattogram
Bangladesh

Situation

Latitude 22.46 °N
Longitude 91.97 °E
Altitude 29 m
Time zone UTC+6

Project settings

Albedo 0.20

Weather data

EME building, CUET, Chattogram
Meteonorm 8.1 (1991-2012), Sat=100% - Synthetic

System summary

Standalone system

PV Field Orientation

Fixed plane
Tilt/Azimuth 22.6 / 0 °

Standalone system with batteries

Near Shadings

Linear shadings : Fast (table)

User's needs

Daily household consumers
Constant over the year
Average 15.4 kWh/Day

System information

PV Array

Nb. of modules 72 units
Pnom total 4320 Wp

Battery pack

Technology Lead-acid, vented, plates
Nb. of units 44 units
Voltage 48 V
Capacity 1100 Ah

Results summary

Useful energy from solar	5257.41 kWh/year	Specific production	1217 kWh/kWp/year	Perf. Ratio PR	70.02 %
Missing Energy	357.75 kWh/year	Available solar energy	5716.65 kWh/year	Solar Fraction SF	93.63 %
Excess (unused)	186.40 kWh/year				

Table of contents

Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
Near shading definition - Iso-shadings diagram	5
Detailed User's needs	6
Main results	7
Loss diagram	8
Predef. graphs	9



Project: EME building Corridor Illumination

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
02/02/26 19:54
with V7.4.7

General parameters

Standalone system

PV Field Orientation

Orientation

Fixed plane
Tilt/Azimuth 22.6 / 0 °

Near Shadings

Linear shadings : Fast (table)

Standalone system with batteries

Sheds configuration

Nb. of sheds 2 units

Sizes

Sheds spacing 38.1 m
Collector width 1.02 m
Ground Cov. Ratio (GCR) 2.7 %
Top inactive band 0.02 m
Bottom inactive band 0.02 m

Shading limit angle

Limit profile angle 0.6 °

User's needs

Daily household consumers
Constant over the year
Average 15.4 kWh/Day

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

PV Array Characteristics

PV module

Manufacturer Generic
Model Poly 60 Wp 36 cells
(Original PVsyst database)

Unit Nom. Power 60 Wp
Number of PV modules 72 units
Nominal (STC) 4320 Wp
Modules 18 string x 4 In series

At operating cond. (50°C)

Pmpp 3891 Wp
U mpp 61 V
I mpp 64 A

Controller

Universal controller
Technology MPPT converter
Temp coeff. -5.0 mV/°C/Elem.

Converter

Maxi and EURO efficiencies 97.0 / 95.0 %

Total PV power

Nominal (STC) 4.32 kWp
Total 72 modules
Module area 40.1 m²
Cell area 33.7 m²

Battery

Manufacturer Generic
Model Open 12V / 100 Ah
Technology Lead-acid, vented, plates
Nb. of units 11 in parallel x 4 in series
Discharging min. SOC 20.0 %
Stored energy 42.2 kWh

Battery Pack Characteristics

Voltage 48 V
Nominal Capacity 1100 Ah (C10)
Temperature Fixed 27 °C

Battery Management control

Threshold commands as SOC calculation
Charging SOC = 0.92 / 0.75
approx. 54.1 / 50.1 V
Discharging SOC = 0.20 / 0.45
approx. 47.2 / 48.9 V

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

Module Quality Loss

Loss Fraction 2.5 %

DC wiring losses

Global array res. 16 mΩ
Loss Fraction 1.5 % at STC

Module mismatch losses

Loss Fraction 2.0 % at MPP

Serie Diode Loss

Voltage drop 0.7 V
Loss Fraction 1.0 % at STC

Strings Mismatch loss

Loss Fraction 0.2 %



PVsyst V7.4.7

VC0, Simulation date:

02/02/26 19:54

with V7.4.7

Array losses

IAM loss factor

ASHRAE Param.: $IAM = 1 - bo (1/\cos i - 1)$

bo Param. 0.05



Project: EME building Corridor Illumination

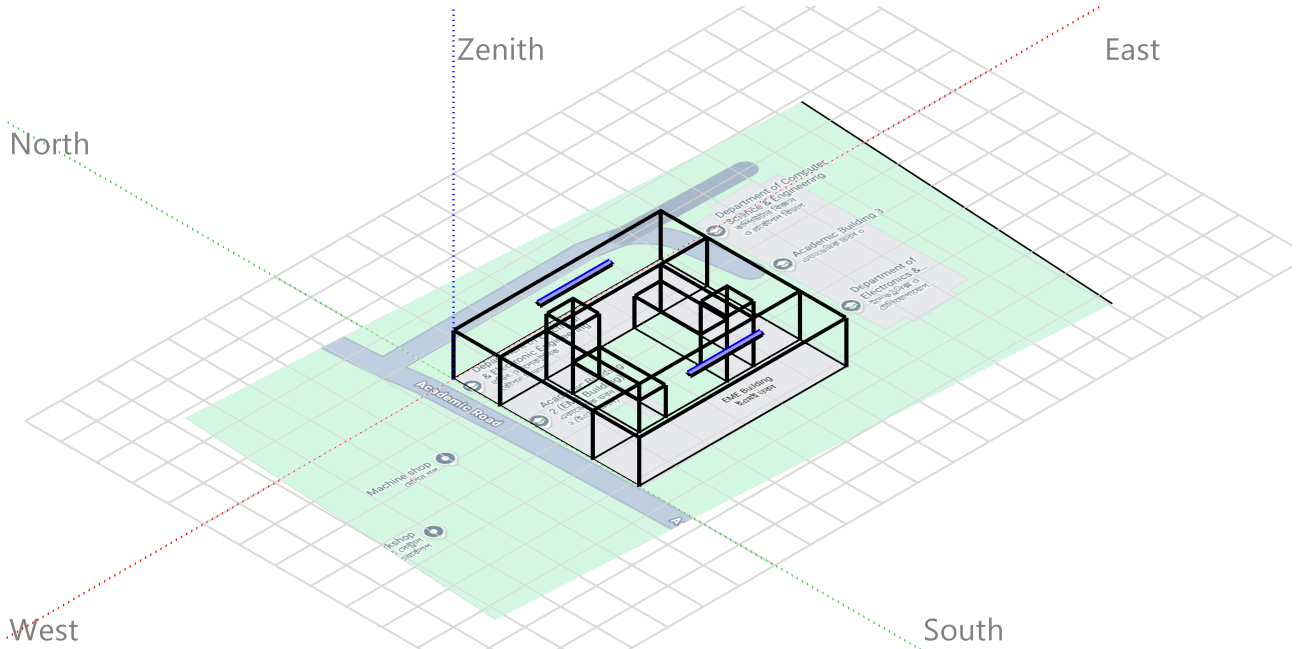
Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
02/02/26 19:54
with V7.4.7

Near shadings parameter

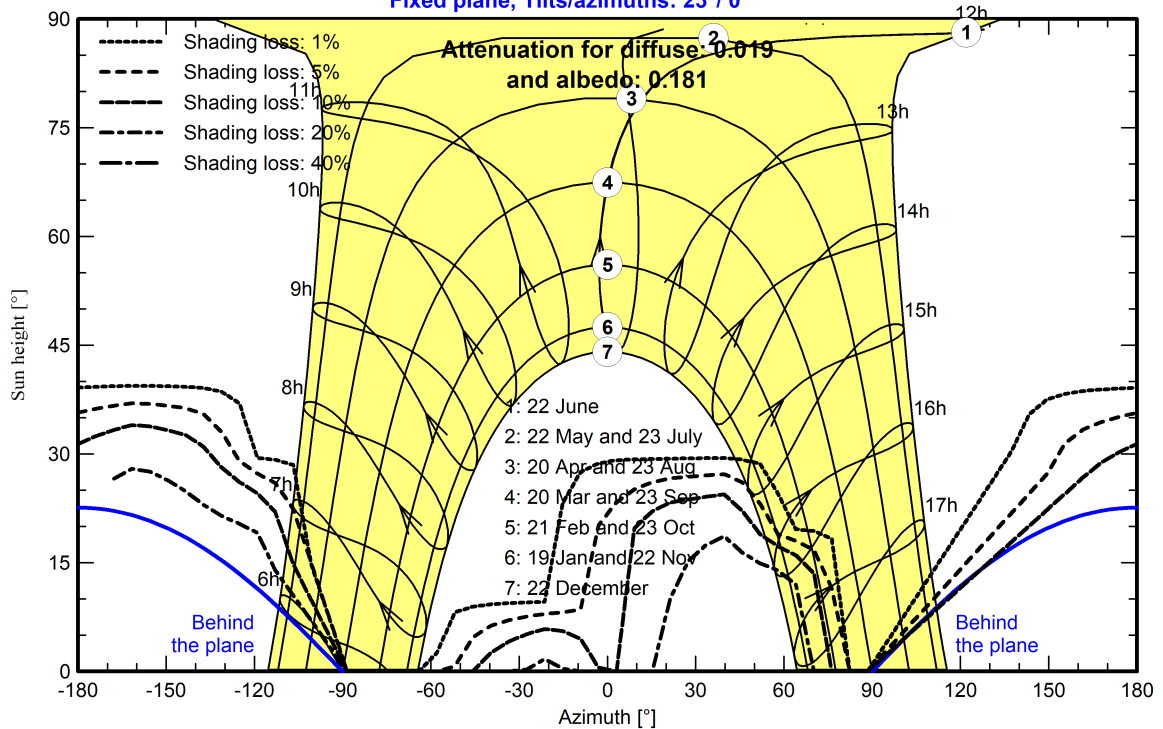
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 23°/ 0°





PVsyst V7.4.7

VC0, Simulation date:
02/02/26 19:54
with V7.4.7

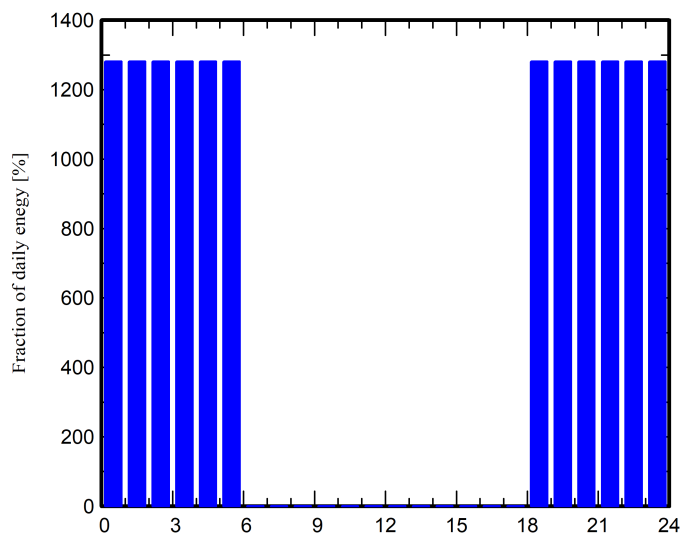
Detailed User's needs

Daily household consumers, Constant over the year, average = 15.4 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	64	20/lamp	12.0	15360
Stand-by consumers			24.0	24
Total daily energy				15384

Hourly distribution





Project: EME building Corridor Illumination

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
02/02/26 19:54
with V7.4.7

Main results

System Production

Useful energy from solar 5257.41 kWh/year
Available solar energy 5716.65 kWh/year
Excess (unused) 186.40 kWh/year

Perf. Ratio PR 70.02 %
Solar Fraction SF 93.63 %

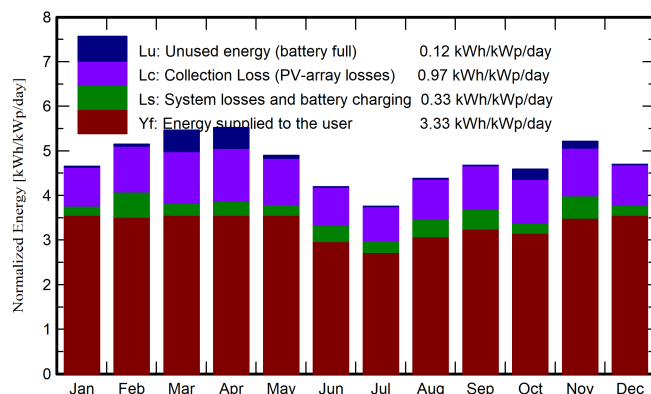
Loss of Load

Time Fraction 0.0 %
Missing Energy 357.75 kWh/year

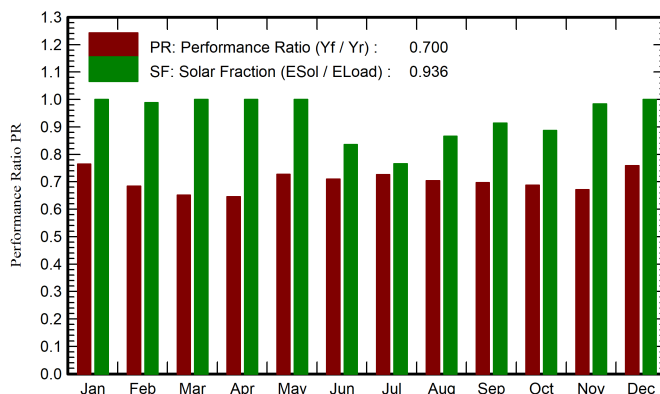
Battery aging (State of Wear)

Cycles SOW 94.2 %
Static SOW 88.5 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m ²	kWh/m ²	kWh	kWh	kWh	kWh	kWh	ratio
January	116.5	139.0	487.6	2.31	0.0	476.9	476.9	1.000
February	123.4	139.9	478.6	4.32	5.1	425.6	430.8	0.988
March	156.9	164.6	553.9	62.67	0.0	476.9	476.9	1.000
April	165.9	160.3	540.7	58.94	0.0	461.5	461.5	1.000
May	160.9	146.1	495.4	8.23	0.0	476.9	476.9	1.000
June	136.6	120.8	411.7	0.06	75.7	385.9	461.5	0.836
July	125.0	111.5	380.2	0.03	111.6	365.3	476.9	0.766
August	140.6	130.7	443.3	1.20	64.1	412.8	476.9	0.866
September	135.7	135.3	458.0	0.00	39.7	421.8	461.5	0.914
October	128.0	137.9	463.6	29.79	53.9	423.1	476.9	0.887
November	125.6	151.5	516.5	18.84	7.7	453.8	461.5	0.983
December	113.0	140.2	487.0	0.01	0.0	476.9	476.9	1.000
Year	1628.1	1677.8	5716.6	186.40	357.7	5257.4	5615.2	0.936

Legends

GlobHor Global horizontal irradiation
GlobEff Effective Global, corr. for IAM and shadings
E_Avail Available Solar Energy
EUnused Unused energy (battery full)
E_Miss Missing energy

E_User Energy supplied to the user
E_Load Energy need of the user (Load)
SolFrac Solar fraction (EUsed / ELoad)



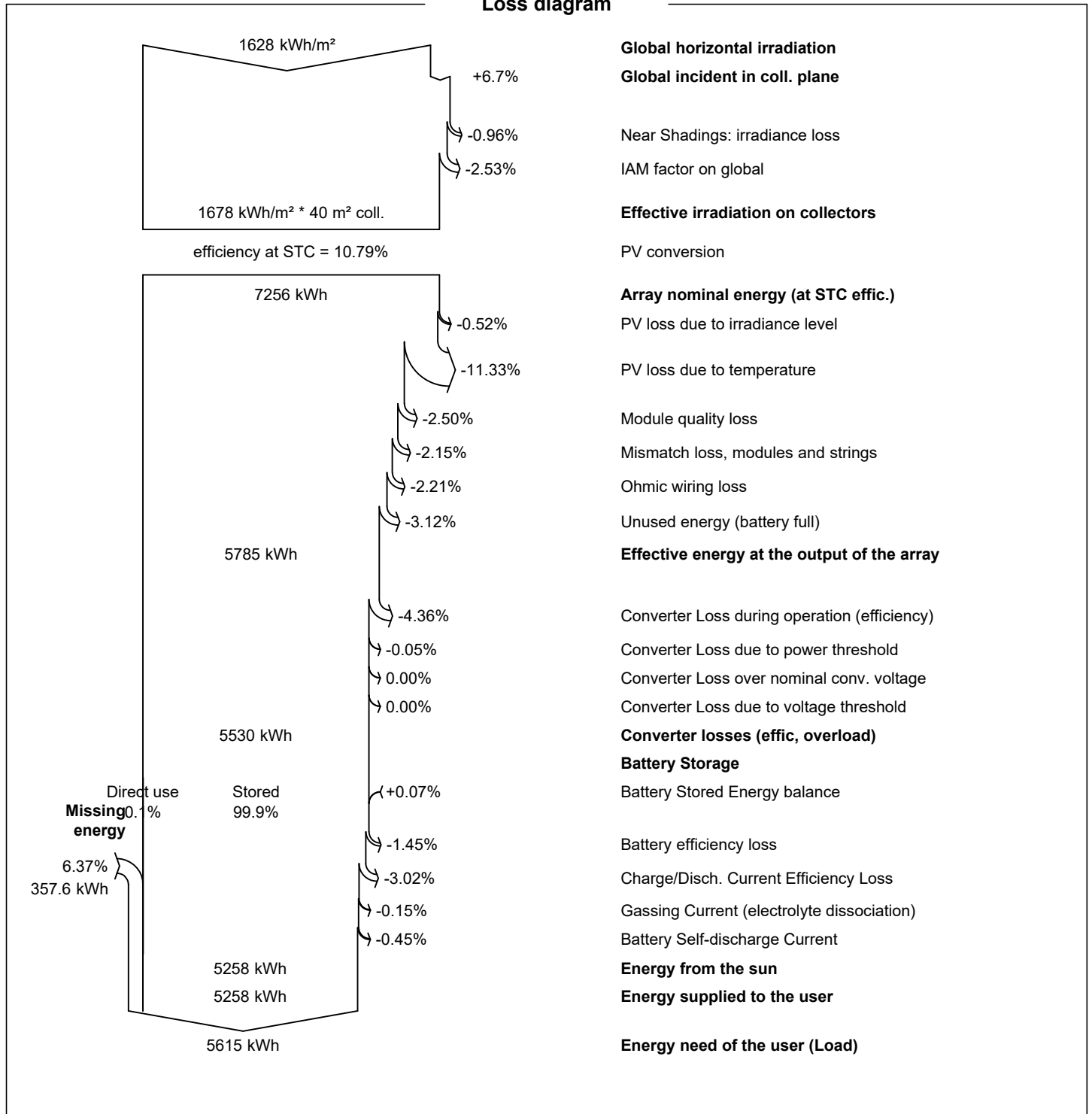
Project: EME building Corridor Illumination

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
02/02/26 19:54
with V7.4.7

Loss diagram





PVsyst V7.4.7

VC0, Simulation date:
02/02/26 19:54
with V7.4.7

Predef. graphs

Daily Input/Output diagram

